01 / Claim

What capability are we testing, under which conditions?

EX06 / NeuralFeels | Source analysis | C1 | P: paper; W: project; R: repository

For the reported unknown-shape SLAM sequences, fusing touch with vision improves pose estimation; it does not establish better control.

1. System and capability

Name your system, tested capability and physical mechanism. Separate your proposed claim from published results.

An Allegro hand rotates objects while NeuralFeels jointly fits shape and pose from RGB-D, DIGIT touch and joint configuration. [P Fig 2; Section 4]

2. Conditions and boundary

State task, configuration, conditions and exclusions. Use Project development for your own final project.

Choose unknown-shape SLAM, not known-CAD tracking. Report physical and simulated sequences separately; the exploration controller is not the contribution being tested. [P 2.2; Fig 3]

3. Measure and comparison

Define measure, units, trial unit, criterion and comparator before evaluation. Label proposed versus observed values.

Use ADD-S pose drift in mm and F-score at 5 mm. Compare vision-only with visuo-tactile inference on the same sequences/reference. [P 2.1; Fig 3]

4. Purpose and next test

Select one feasible pilot and check resources/access. State what outcome would challenge the claim; results may be pending.

Offline pilot: extract Fig 3 real/sim contrasts and reference construction. Pass: metric, modality, sequence count and estimation seeds are separately recorded.

Review: Could your system work as shown while your claim is still false? Write one such outcome.

Example response: An improved pose estimate is not itself improved manipulation; keep perception as the claimed outcome.

Worksheet 1 / Claim – evidence and team review

EX06 / NeuralFeels | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX06 Project: NeuralFeels Claim ID: C1 Revision: draft 2

Mode: Source analysis Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: selected paper text/tables checked 20 Oct 2026; no raw dataset download or independent experiment.

Evidence ID	Artifact/source + location (clickable)	What it supports; limitation / access gap
P	arxiv.org/html/2312.13469v1	Paper: method/results; reported evidence, not our measurement.
W	suddhu.github.io/neural-feels	Project: system/videos; selected examples do not establish reliability.
R	github.com/facebookresearch/neuralfeels	Repository: linked route; code, raw data and current commit not audited.

Next action / owner / verification:

Analyst: record Fig 3 metrics, 40/30 sequence counts and five seeds; separate unknown-shape from known-CAD tests.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

2. Name / student ID: _____

Signature: _____ Date: _____

3. Name / student ID: _____

Signature: _____ Date: _____

4. Name / student ID: _____

Signature: _____ Date: _____

5. Name / student ID: _____

Signature: _____ Date: _____

6. Name / student ID: _____

Signature: _____ Date: _____

7. Name / student ID: _____

Signature: _____ Date: _____

8. Name / student ID: _____

Signature: _____ Date: _____

P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.